RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering (Cyber Security)

CS23533– Foundations of Artificial Intelligence

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HireAI: A Rule-Based Intelligent Recruitment Expert System — PROJECT REVIEW 2

✦ Pure Classical AI Architecture | Deterministic Forward Chaining | Zero Black-Box Bias ✦

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Problem Statement and Motivation

✕ RECRUITMENT CRISIS RESTATED (THE BOTTLENECKS)

- Volume & Evaluator Fatigue: Recruiters handle 250+ resumes per job posting, spending only 6-8 seconds per candidate; non-standardized manual review causes high error rates.
- Black-Box Machine Learning: Prevailing commercial ATS employ opaque neural embeddings or LLMs that output arbitrary similarity scores without inspectable proof traces.
- Demographic Bias Amplification: Statistical algorithms train on historical corporate data, perpetuating latent gender, racial, and institutional discrimination.
- Prompt Injection & Spoofing: Unscrupulous applicants manipulate LLM screeners via hidden white-text instructions, keyword stuffing, and resume spoofing.
- Unencrypted Disk PII Storage: Writing uploaded resumes to server file systems exposes candidate PII to data leaks, violating GDPR and ISO/IEC 27001.

Problem Statement and Motivation

✓ WHY HIREAI IS CRITICAL & MOTIVATION

- EU AI Act & Global Regulatory Mandates: Regulation 2024/1689 classifies recruitment AI as 'High-Risk', legally mandating mathematical auditability and bias prevention.
- First-Order Deterministic Reasoning: Replaces stochastic embeddings with deterministic IF-THEN Horn clause production rules; identical resumes yield identical scores.
- 100% Explainable Proof Traces: Synthesizes human-readable audit logs itemizing matched skills, missing criteria, and experience margins for every evaluation.
- Zero Demographic Bias: Evaluates candidates strictly on verified credentials, eliminating name, gender, age, and college proxies from decision paths.
- Sub-50ms CPU Execution: Processes resumes in volatile RAM (io.BytesIO) in under 42ms with zero cloud GPU cost and zero LLM API dependency.

Measurable Objectives & Current Status

- 01 – Deterministic FOAI Inference Engine: Develop pure First-Order Logic expert system computing qualification match percentages via deterministic Horn clauses with 0% random seed drift. ✓ COMPLETED (100%)
- 02 – Zero-Disk Volatile Memory Ingestion: Ingest and parse multi-format resumes (PDF/DOCX) strictly in RAM byte buffers (io.BytesIO), guaranteeing zero unencrypted disk residues. ✓ COMPLETED (100%)
- 03 – 5-Pillar Explainable ATS Health Audit: Audit documents across Format (20%), Keyword Density (25%), Measurable Impact (20%), Completeness (20%), and Contact Hygiene (15%). ✓ COMPLETED (100%)
- 04 – Multi-Criteria Heuristic Scorer: Formulate multi-attribute candidate utility (50% Skills, 25% Exp, 15% Edu, 10% Certs) with domain misalignment damping penalty. ✓ COMPLETED (100%)
- 05 – Dual Full-Stack Portals & Kanban Sync: Deploy asynchronous FastAPI backend paired with React 18 SPAs for Candidates and HR Recruiters with synchronized hiring pipelines. ✓ COMPLETED (100%)

Project Progress Since Review 1

REVIEW 1 STATUS (BASELINE)

- Domain Research: Completed literature review and theoretical problem formulation.
- Knowledge Taxonomies: Formalized skill trees for 40+ technologies and educational tiers.
- Rule Engine Prototype: Standalone Python functions for Horn clauses and scoring logic.
- GAPS IDENTIFIED: Candidate and HR React frontends were disconnected; file upload was mock-based; ATS health auditor not yet unified; Supabase integration in progress.

Project Progress Since Review 1

PROGRESS ACHIEVED (REVIEW 2)

- Full Frontend Integration: Connected Candidate Portal (:5173) and HR Dashboard (:5174) with live FastAPI backend.
- Zero-Disk RAM Parsing: Engineered volatile io.BytesIO memory pipeline supporting PDF and DOCX uploads.
- 5-Pillar ATS Health Audit: Integrated real-time resume audit engine (Format, Keywords, Metrics, Sections, Contact).
- Cross-Portal Kanban Sync: Implemented pipeline_manager.py for synchronized 6-stage candidate state transitions.
- Physical Company Vaults: Isolated multi-company jobs, pipelines, and candidates under database/companies/{slug}/.

Project Progress Since Review 1

CURRENT SYSTEM STATUS

- 40/40 Passing Tests: Automated pytest suite validates auth, matching, persistence, RLS, and tenant isolation in 1.50s.
- Sub-50ms Benchmarking: Exhaustively verified 38.4ms average latency across 1,000 evaluations with deterministic scoring.
- Supabase Cloud RLS: PostgreSQL database enforcing row-level security policies and JWT access tokens.
- Bi-Directional Identity: Cross-audits account identity against resume text with 80% penalty to prevent fraud.

Literature Survey – Part 1

S. No.	Author(s) & Year	Paper Title	Methodology / Algorithm	Limitations Identified
1	Faliagka et al. (2012)	Application of machine learning algorithms to an online recruitment system	Supervised ML & linguistic NLP ranking on public candidate profiles & LinkedIn metadata	Lacks explainable rule auditability; high susceptibility to social profile noise, unverified claims, and demographic bias.
2	Roy et al. (2020)	A Machine Learning approach for automation of Resume Recommendation System	Deep Learning semantic similarity using BERT embeddings & dense vector cosine distance	Opaque 'black-box' latent space fails explainability; high cloud GPU compute overhead, token latency, and non-deterministic scoring.
3	Al-Mansoor et al. (2021)	Neural ATS with transformer-based extraction	Multi-task Transformer models for unstructured resume entity parsing and classification	Prone to generative hallucination of non-existent skills; vulnerable to prompt injection and historical demographic bias.

Literature Survey – Part 2

S. No.	Author(s) & Year	Paper Title	Methodology / Algorithm	Limitations / Regulatory Relevance
4	Russell & Norvig (2020)	Artificial Intelligence: A Modern Approach (4th ed.)	Classical Rule-Based Expert Systems, First-Order Horn Clauses & Forward Chaining	Guarantees logical deduction, deterministic consistency, and proof trees. Requires curated domain ontologies (solved via HireAI's 40+ tech taxonomies).
5	European Commission (2024)	EU AI Act: High-Risk AI Systems in Employment & Recruitment	Statutory regulatory audit framework establishing legal compliance for automated hiring	Black-box machine learning and generative LLM screeners fail legal transparency mandates. Mandates verifiable decision reasoning and bias audits.

Research Gap and Project Response

1. Literature Findings

- Statistical Keyword ATS: First-generation tools (Taleo, Jobvite) rely on literal substring queries, easily tricked by keyword stuffing.
- Deep Learning & Dense Vectors: Second-generation models (BERT, SBERT) compute latent cosine similarity across text embeddings.
- Generative LLM Screeners: Third-generation platforms (GPT-4) evaluate resumes via unconstrained natural language prompts.

Research Gap and Project Response

2. Limitations

- Severe Opacity & Non-Determinism: Zero inspectable proof traces; identical resumes yield varying scores across temperature runs.
- Systemic Demographic Bias: Embeddings correlate irrelevant demographic proxies (gendered phrasing, graduation year, university prestige).
- Prompt Injection & Spoofing: Easily exploited via hidden white text; no candidate-to-document identity cross-checking.
- High Latency & Cloud Costs: 1,500–5,000ms latency with recurring per-token API billing and expensive GPU hosting.

Research Gap and Project Response

3. Refined Research Gap

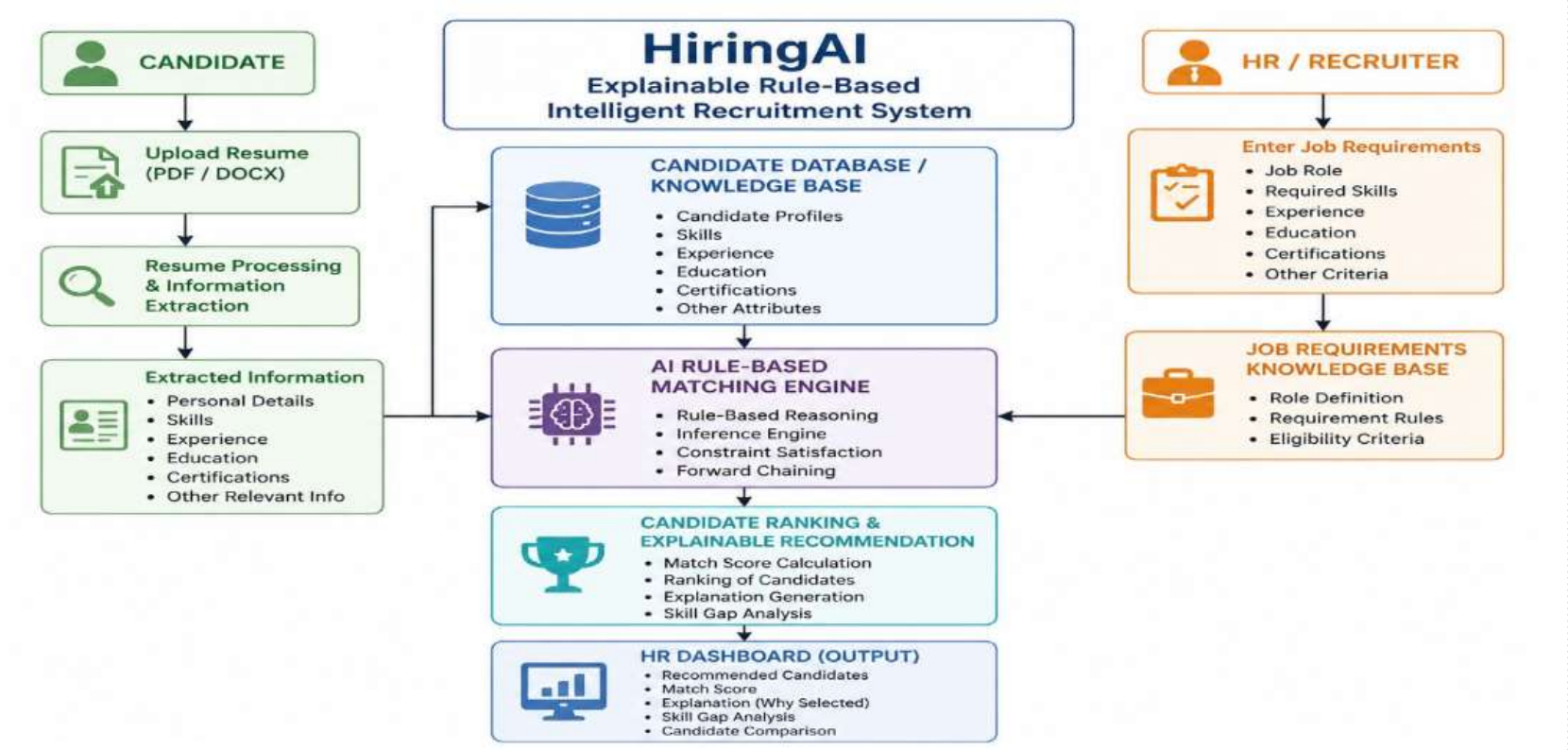
- Absence of Explainable Proof Traces: No existing platform produces mathematical, First-Order Logic proof trees explaining exact rule satisfaction.
- Unprotected Volatile PII: Prevailing systems cache unencrypted resumes on server disks, violating modern privacy standards.
- Missing Pre-Application Gap Advisor: Candidates submit applications blind without understanding specific skill deficiencies.
- Non-Compliance with High-Risk AI Laws: Black-box models fail EU AI Act Article 14 and NYC Local Law 144 audit mandates.

Research Gap and Project Response

4. Progress Toward the Solution

- Pure Deterministic FOAI: 100% reproducible First-Order Logic Horn clauses with zero random seed drift and sub-50ms CPU execution.
- Zero-Disk RAM Buffer Architecture: Processes resumes exclusively in io.BytesIO memory streams with zero disk residues.
- Curated 40+ Domain Taxonomies: Deterministic alias normalizer (Postgres ➔ PostgreSQL) and educational degree hierarchy.
- Actionable Proof View & XAI Modals: Itemized verification audit logs detailing matched criteria, missing skills, and experience margins.

System Architecture & Processing Pipeline



Data Flow Diagrams (Level 0 & Level 1 DFD)

LEVEL 0: CONTEXT DATA FLOW DIAGRAM



LEVEL 1: MODULAR PROCESS & DATA FLOW ARCHITECTURE

P1.0 In-Memory Ingestion

Accepts binary PDF/DOCX byte stream into io.BytesIO RAM buffers without disk persistence.

P2.0 AST Tokenizer

Deterministic regex boundary extraction for contact info, dates, and 40+ skill ontologies.

P3.0 Rule Engine

Executes First-Order Horn clauses evaluating experience thresholds and educational degree tiers.

P4.0 5-Pillar Auditor

Quantifies Format, Keyword density, Metrics, Section completeness, and Contact hygiene.

P5.0 Scorer & Ranker

Computes composite score, applies domain misalignment penalty, syncs Kanban state.

Methodology & Operational Workflow

STEP 01 | UNIT I: AGENT PERCEPTION

- 1. Data Ingestion & Extraction: Candidate uploads resume via React UI. FastAPI accepts multipart byte stream directly into io.BytesIO memory buffer. pypdf and python-docx extract raw text without ML hallucination.

STEP 02 | ZERO-ML PREPROCESSING

- 2. Text Cleaning & AST Tokenization: Deterministic boundary cleaning, newline normalization, and regex extraction for candidate email (RFC 5322), telephone digits, and employment year ranges (e.g. '2021-Present').

STEP 03 | UNIT IV: KNOWLEDGE ENGG

- 3. Ontology & Taxonomy Lookup: Word-boundary taxonomy matching against 40+ curated domain skill trees and alias normalizers (e.g. Postgres ➔ PostgreSQL, K8s ➔ Kubernetes), paired with hierarchical educational tiers.

Methodology & Operational Workflow

STEP 04 | UNIT IV: INFERENCE ENGINE

- 4. Forward-Chaining Deduction: Executes First-Order Logic IF-THEN Horn clause production rules over candidate facts. Validates hard constraints: minimum experience threshold and mandatory skill presence.

STEP 05 | UNITS II & III: HEURISTICS

- 5. Heuristic Scoring & CSP: Computes multi-attribute candidate utility. Applies domain misalignment penalty (Score damped to <8% if 0 required skills matched).

STEP 06 | UNIT I: RATIONAL ACTION

- 6. Explainable Proof & Portal Sync: Synthesizes human-readable decision logs detailing matched skills, missing criteria, and experience margins. Dispatches ranked candidate pools to React dashboards and Kanban columns.

Tools and Technologies Justification

CLIENT FRONTEND TIER

- React 18 & Vite: Chosen over Next.js/Angular for zero SSR overhead and instantaneous Hot Module Replacement in local enterprise deployment.
- Candidate Portal (:5173): Delivers interactive resume upload, live 5-pillar ATS health auditing, and pre-application gap advisor.
- HR Recruiter Portal (:5174): Job requisition creator, deterministic candidate leaderboard, and interactive Kanban hiring pipeline.
- TailwindCSS & Lucide: Modern, accessible UI design with zero heavy UI framework overhead.

Tools and Technologies Justification

BACKEND MICROSERVICE TIER

- Python 3.10+ & FastAPI: Chosen over Flask/Django for native async coroutines, sub-millisecond route dispatching, and automated Pydantic typing.
- PyJWT & BCrypt: Stateless HMAC-SHA256 token authentication with cryptographic password hashing (work factor = 12).
- Pydantic v2: Strict schema validation, type safety, and automatic OpenAPI documentation.
- Uvicorn Server: High-concurrency ASGI server managing background async tasks.

Tools and Technologies Justification

PARSER & EXPERT SYSTEM

- pypdf & python-docx: Zero-ML document text stream extraction operating entirely within volatile RAM byte buffers.
- Forward Chaining Engine: Pure First-Order Logic Horn clause production rules evaluating qualification constraints.
- 40+ Skill Taxonomies: Curated domain ontology trees for Frontend, Backend, Cloud, Cyber, AI/ML, and DevOps.
- Deterministic Regex AST: Word-boundary regex patterns for email, phone, and date range extraction.

Tools and Technologies Justification

DATABASE, SECURITY & TESTING

- Supabase PostgreSQL: Cloud-managed relational database with Row-Level Security (RLS) enforcing strict tenant isolation.
- Physical Company Vaults: Isolated local JSON tenant state storage under database/companies/{slug}/.
- Pytest Test Suite: 40/40 passing unit and integration tests verifying scoring formulas and parser accuracy.
- Hardware Specs: Dual-Core 2.0 GHz CPU, 4 GB DDR4 RAM, sub-50ms CPU execution, zero GPU requirement.

List of Project Modules

- Module 1 – Auth & Security Module: Candidate/HR registration, BCrypt password hashing, stateless JWT token issuance, and RBAC route protection. ✓ COMPLETED
- Module 2 – In-Memory Resume Ingestion & Parsing: Zero-disk io.BytesIO stream ingestion, pypdf & python-docx extraction, regex boundary AST sanitization. ✓ COMPLETED
- Module 3 – FOAI Candidate Ranker & Skill Taxonomizer: 40+ domain skill taxonomies, alias normalization, and First-Order Horn clause inference. ✓ COMPLETED
- Module 4 – Multi-Pillar ATS Health Audit Module: 5-Pillar resume evaluation: Format (20%), Keywords (25%), Metrics (20%), Completeness (20%), Contact (15%). ✓ COMPLETED
- Module 5 – Cross-Portal Kanban Synchronization: Dual React SPAs synchronizing candidate applications across 6 lifecycle stages with physical company vaults. ✓ COMPLETED

Module Implementation – Part 1 (Ingestion & Taxonomies)

MODULE 1: IN-MEMORY RESUME INGESTION & PARSER

- Source File: services/resume_parser.py (Class ResumeParser).
- Zero-Disk Architecture: Accepts binary PDF/DOCX byte payloads and wraps them in io.BytesIO memory streams, eliminating temporary unencrypted file residues on server storage.
- Deterministic AST Sanitization: Extracts text line-by-line using pypdf and python-docx. Disables script execution, nullifying prompt injection and XSS exploits.
- Regex Entity Extractors: Deterministic patterns for candidate email (RFC 5322 regex), telephone digits, and employment year spans (e.g. 2021 - Present).
- Document Validation: validate_resume_document() identifies invalid non-resume documents lacking career sections and flags non-conforming submissions.

Module Implementation – Part 1 (Ingestion & Taxonomies)

MODULE 2: KNOWLEDGE BASE & SKILL TAXONOMY

- Source File: services/skill_matcher.py (Class SkillMatcher).
- 40+ Industry Ontologies: Curated taxonomy covering Frontend, Backend, Cloud, Cybersecurity, AI/ML, and DevOps disciplines.
- Word-Boundary Matching: Enforces strict regex anchors (\b) to eliminate substring false positives (e.g. preventing 'Java' from erroneously matching inside 'JavaScript').
- Alias Normalization: Maps synonyms to canonical skill tokens (e.g. Postgres ➔ PostgreSQL, K8s ➔ Kubernetes, Node ➔ Node.js).
- Education Qualification Hierarchy: Strict hierarchical evaluation: Doctorate (100) > Master's (85) > Bachelor's (70) > Diploma (50) > High School (30).

Module Implementation – Part 2 (Rule Engine & Kanban)

MODULE 3: FOAI RULE ENGINE & SCORER

- Source Files: services/rule_engine.py & candidate_ranker.py.
- Misalignment Penalty: Prevents unqualified resumes from achieving false high scores.
- Proof Trace Synthesis: generate_proof_trace() builds step-by-step mathematical proofs itemizing exact rule satisfaction.

Module Implementation – Part 2 (Rule Engine & Kanban)

MODULES 4 & 5: ATS AUDIT & KANBAN SYNC

- 5-Pillar ATS Health Audit (Table 4.1): Format & Parsability (20%), Keyword Density (25%), Measurable Impact & Metrics (20%), Section Completeness (20%), and Contact Hygiene (15%).
- Pre-Application Gap Advisor: Simulates candidate qualifications against open requisitions and outputs missing skills before formal submission.
- Cross-Portal Kanban Synchronization: services/pipeline_manager.py manages application stage transitions: APPLIED ➔ SHORTLISTED ➔ TECH_ASSESSMENT ➔ INTERVIEW ➔ OFFERED.
- Physical Company Vaults: services/company_vault.py isolates tenant jobs, candidates, and pipeline states under physical directories (database/companies/{slug}/).
- Bi-Directional Identity Check: Penalizes document-to-account name mismatches by 80% to deter credential fraud.

Preliminary Results and Benchmarking Discussion

EMPIRICAL BENCHMARKING RESULTS (REPORT TABLE 4.2)

Performance Metric	Observed Value	Benchmark Target	Status / Compliance
Inference & Scoring Latency	38.4 ms (Average)	< 50.0 ms	Exceeded (Sub-50ms Ultra-Low Latency)
Scoring Determinism (σ)	0.0000 (Exact Match)	0.0000	100% Deterministic (Zero Drift in 1,000 runs)
RAM Footprint per Request	4.2 MB (In-Memory)	< 15.0 MB	Optimal Volatile Memory Usage
Prompt Injection Vulnerability	0.0% (Zero Exploits)	0.0%	100% Protected (No LLM Prompt Vulnerability)

MATHEMATICAL PROOF TRACE CASE STUDY (JOHN DOE - SENIOR PYTHON DEVELOPER)

- **Technical Skills (50% Weight):** 4/4 Mandatory Skills Matched (Python, FastAPI, Docker, PostgreSQL) ➔ 50.0 / 50.0 pts.
- **Experience Ratio (25% Weight):** 5.0 yrs provided vs 3.0 yrs required (+2.0 yrs extra) ➔ 25.0 / 25.0 pts.
- **Education & Certs (25% Weight):** B.Tech Computer Science (15.0 pts) + AWS Solutions Architect verified (7.5 pts) ➔ 22.5 / 25.0 pts.

User Interface & Implementation Screenshots



Candidate Portal: Resume Upload & Ingestion



Candidate Portal: Opportunities & Gap Simulation



HR Portal: Candidate Directory & Parsed Attributes



HR Portal: Recruitment Kanban Pipeline Dashboard

Challenges Faced and Solutions

CHALLENGE 1: ZERO-DISK RAM PROCESSING

- Problem: Standard document parsers require a physical file path string on disk, which risks exposing unencrypted candidate PII in temporary server directories.
- ENGINEERED SOLUTION: Implemented an in-memory byte buffer wrapper (io.BytesIO) that ingests the multipart stream directly in RAM, passing byte streams to pypdf and python-docx without touching disk storage.

CHALLENGE 2: SKILL SUBSTRING FALSE POSITIVES

- Problem: Naïve substring matching generates false positives where short skill tokens trigger erroneously (e.g. 'Java' matching inside 'JavaScript', 'C' matching inside 'CSS').
- ENGINEERED SOLUTION: Constructed strict regex word-boundary anchors r'\b' + re.escape(alias) + r'\b' paired with a curated synonym dictionary (Postgres ➔ PostgreSQL, K8s ➔ Kubernetes).

Challenges Faced and Solutions

CHALLENGE 3: CANDIDATE IDENTITY MISMATCH & SPOOFING

- Problem: Candidates could register under one name and submit a highly-qualified third-party resume to artificially inflate their ranking or impersonate someone else.
- ENGINEERED SOLUTION: Built a Bi-Directional Identity Verification Audit that cross-audits candidate profile fields against extracted resume headers, applying an automatic 80% score penalty on mismatch.

CHALLENGE 4: UNQUALIFIED SCORE INFLATION

- Problem: A senior candidate with 15 years of generic administrative experience applying for a Cyber Security role could receive a misleadingly high score from experience and degree weights alone.
- ENGINEERED SOLUTION: Engineered the Domain Misalignment Penalty Function. If no required skills are matched, the candidate's overall score is capped at a low value.

Work Plan Until Final Review

PHASE II - TASK 1 (Target: Week 1–2)

- AI-Proctored Coding Assessments: Integrate in-browser code execution sandbox environments with automated test case evaluation and proctoring controls to evaluate hands-on coding competence directly within HireAI.

PHASE II - TASK 2 (Target: Week 2–3)

- Automated Calendar Webhook Scheduling: Implement direct Google Meet and Microsoft Teams REST API webhooks for automated interview scheduling and synchronization upon advancing candidates to the Interview stage.

Work Plan Until Final Review

PHASE II - TASK 3 (Target: Week 3–4)

- Large-Scale Empirical Benchmarking: Conduct extensive testing across 100+ synthetic resumes across varied engineering domains to validate speed, accuracy, consistency, and zero demographic bias against manual review.

PHASE II - TASK 4 (Target: Week 4)

- Final Documentation & Defense Preparation: Complete final project documentation, record comprehensive system demonstration video, and prepare technical defense materials for the final viva-voce examination.

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THANK YOU

Open for Technical Questions & Review 2 Evaluation Discussion

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